Project II Report

How have the primary concerns of the American people changed over the years, as reflected in the annual Address on the State of the Union delivered by the President?

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ENGDATA101 Introduction to Data Science

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December 9, 2024

The Research Question

The goal is to identify issues that matter to Americans in each passing year. The State of the Union Address is a speech given by the President of the United States to a joint session of Congress. It usually happens at the beginning of the year. The president gives information about world affairs, politics, economics, important issues concerning America, and his plans to address them. The issues discussed are regarded as highly important to the nation, prompting the President to talk about them.

The Data

The data is the transcripts of the State of the Union Addresses available at https://www.presidency.ucsb.edu. Each transcript has its own page but the links do not perfectly follow an organized pattern. Because of that, I obtain the list of links from the general State of the Union Addresses page:

https://www.presidency.ucsb.edu/documents/presidential-documents-archive-guidebook/annual-messages-congress-the-state-the-union

The links are also provided in a table form at:

https://www.presidency.ucsb.edu/documents/presidential-documents-archive-guidebook/annual-messages-congress-the-state-the-union#Table%20of%20SOTU

Processing the data

After I had obtained the links to each speech, I read the page, extracted the year and the name of the president. Other people speaking during the Address are also included in the transcript, their names in italics, so the formatting helps find those lines. I removed them. I also removed all the comments (in italics and square brackets), so only the president's lines remain. I split the speech into individual words and then removed all the stop words (words

that are commonly used but do not provide much information) and numbers. I calculate how many times a word appears in a given speech. Then I merge the data from the speeches delivered between 1997 and 2024.

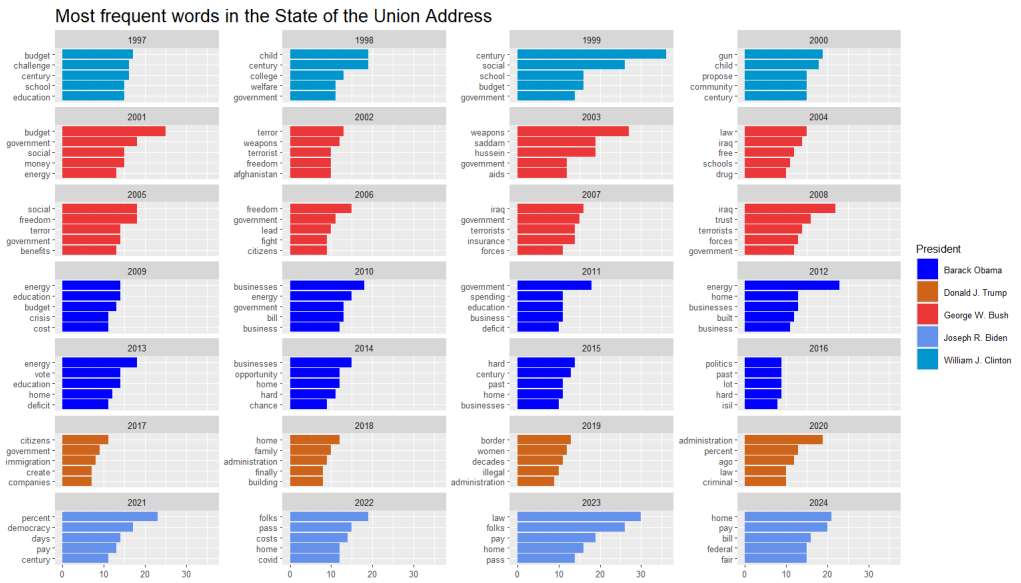
My main goal was to visualize the most common words in each of the State of the Union Addresses. To find the most meaningful words I had to remove those that were too popular and would make all the results seem similar. I decided to remove words that appeared in every one of the 28 considered speeches.

I have also created bar graphs for the more interesting ones so that they are not overlooked. Additionally, I plotted the frequency of some other words that could provide valuable visualizations.

Visualizations

Top 5 words

To visualize the most frequent words in the State of the Union Address each year, I have used a bar graph (geom_col) faceted by year. The color corresponds to the political affiliation of each president. For each year the plot shows 5 most common meaningful words in the speech.



While there are some interesting observations (*century* for Clinton; *terror* and related words for Bush, *covid* in 2022), not very meaningful words (2016: *lot*; 2020: *ago*; 2022 and 2023: *folks*, etc.) still appear in the visualization. It would be difficult to automatically filter them out because that would require an arbitrary measure of meaningfulness.

America

America is the most popular of the words that appear in every State of the Union Address from 1997 to 2024.

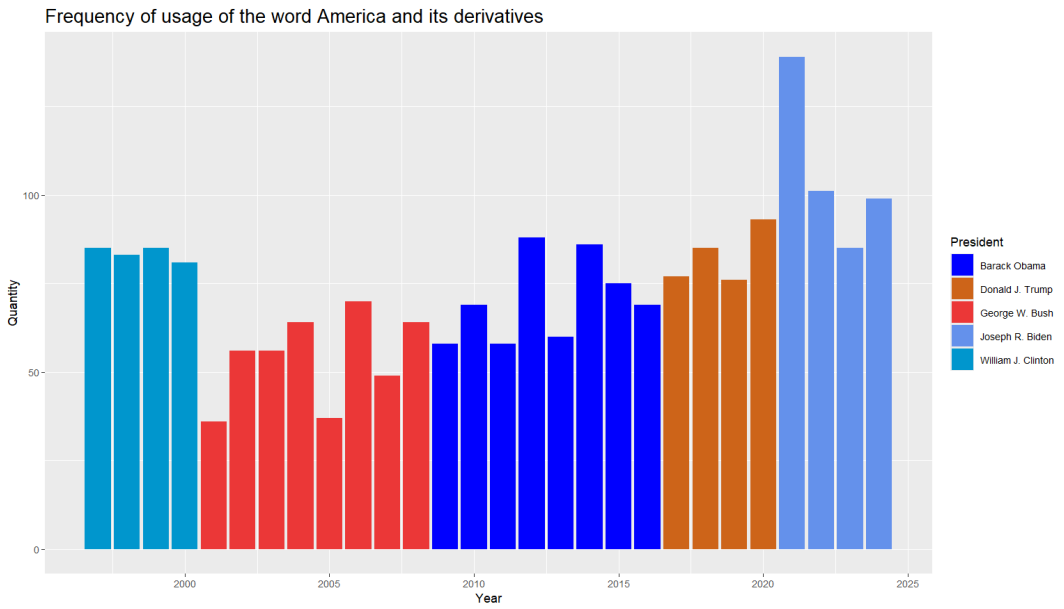
projectII.R popularity

Filter

	word	n	total
1	america	28	797
2	people	28	685
3	american	28	636
4	americans	28	507
5	world	28	453

Its derivatives (*American, Americans*, etc.) are also very common, so I decided to plot the word *America* with its derivatives found using a regular expression *"america[\\w]{0,}"*.

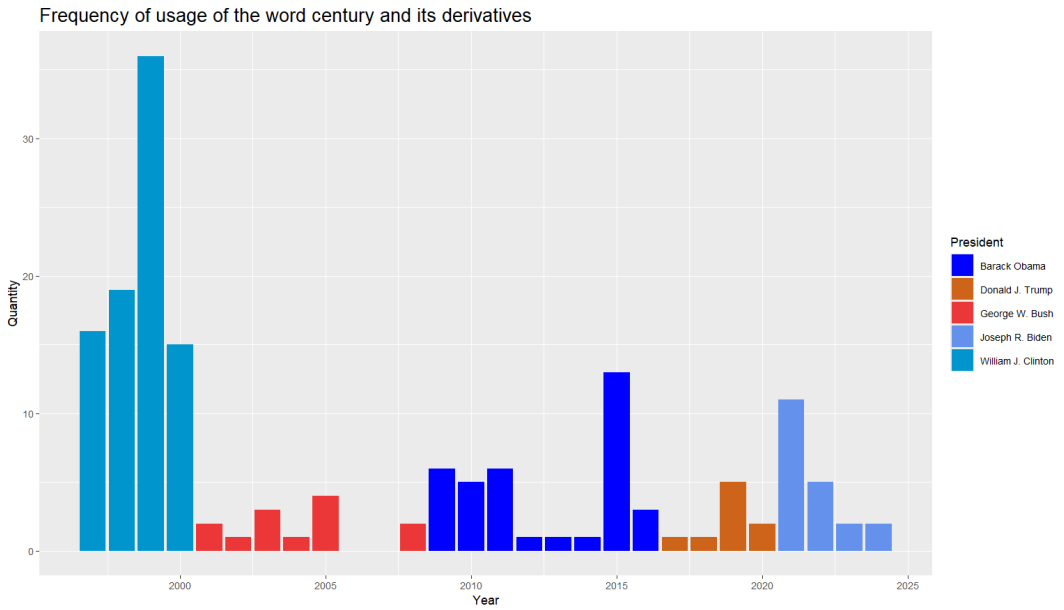
The resulting plot looks as follows:



The plot shows a decline at the beginning of the presidency of George W. Bush. In general, the trend is upward. An unusual peak can be observed in 2021 when Joseph Biden delivered his first Address. Yet, not much about American patriotism can be concluded from this visualization.

Century

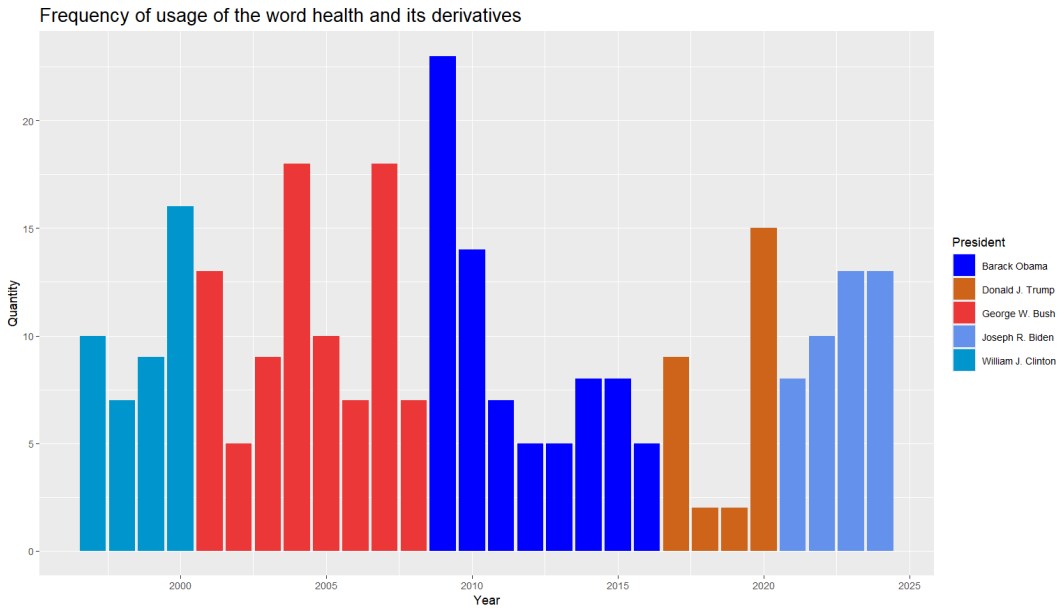
I decided to check the frequency of the word *century* (regular expression: *"centur(ies|y)"*) as it was very popular in Clinton's speeches.



The result reflects the potential anxiety and excitement associated with entering the new century, particularly noticeable in 1999.

Health

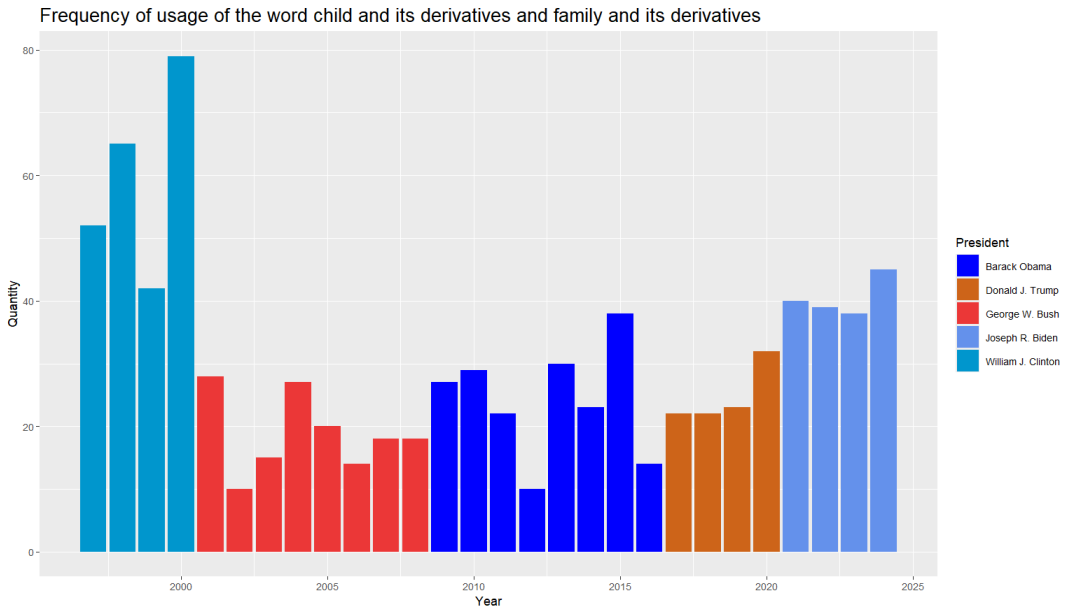
Health-related words (regular expression: *"health[\\w]{0,}"*) seemed interesting because of the pandemic that started in 2020.



Surprisingly, there was no drastic change accompanying the outbreak of the pandemic. The sudden rise in 2020 is not related to the pandemic (yet), there is only one mention of the coronavirus in the 2020 speech. The frequency at which health-related words have been used since 2021 does not differ much from the pre-pandemic average. The sudden spike in 2009 was caused by Barack Obama's proposal for healthcare reform announced in the 2009 Address on the State of the Union.

Child, family

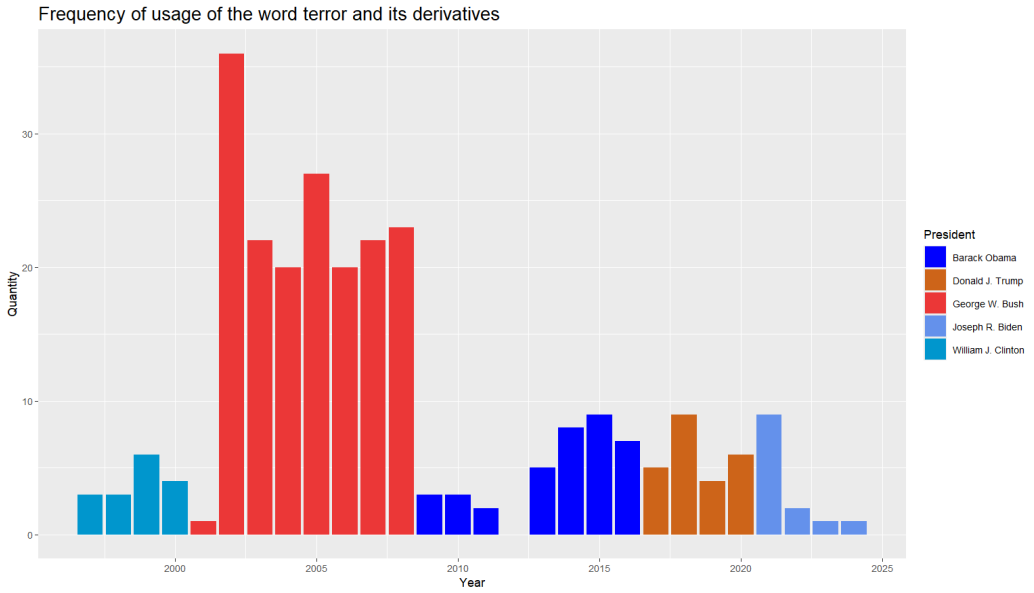
Words related to children and family (found using a regular expression *"(child[\\w]{0,})|(famil(ies|y))"*) appear in every speech.



It seems that the topic of family life was much more important to Americans before 2001.

Terror

While *terror* (and its derivatives found with regular expression "*terror[\\w]{0,}*") does not appear in every speech, the visualization provides an interesting insight.



It clearly shows when terrorism became a serious concern for Americans. Looking at the plot one could conclude it happened in 2002. The State of the Union is delivered at the beginning

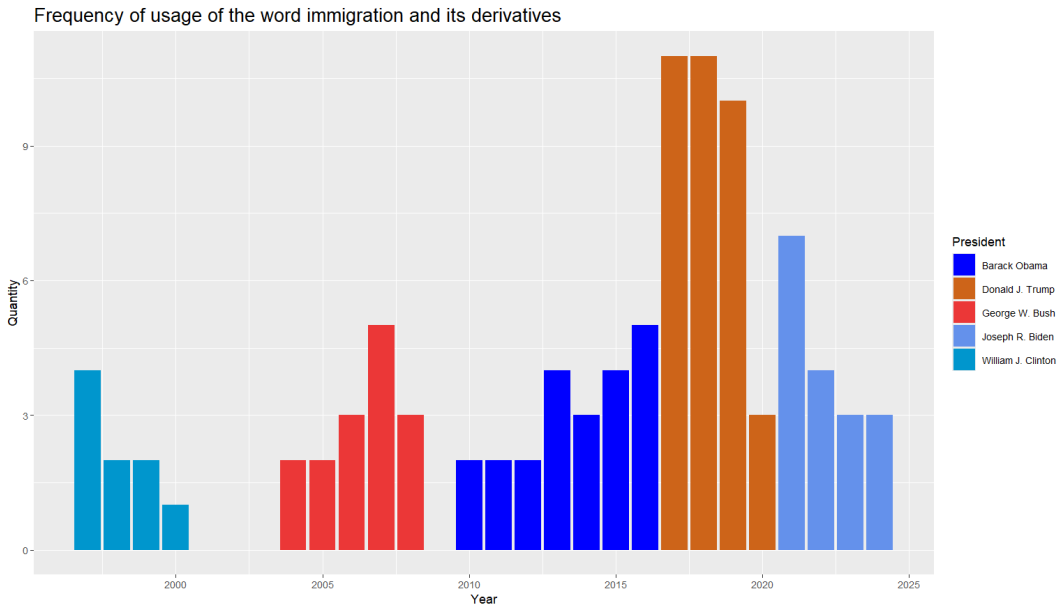
of each year. While the actual reason for the rise in terrorism-related anxiety is the September 11 Attacks from 2001, the 2001 speech does not account for them.

The visualization also shows how the word became a bit less frequent in 2003 and stayed at roughly the same level until the end of Bush's presidency.

The great decline in frequency between 2009 and 2010 suggests that something significant has happened. A possible explanation for such a drastic change is a change in the person delivering the speech. Barack Obama became president in 2010 and possibly decided to change the narrative surrounding terror and America. This shows the disadvantage of using one person's speeches to illustrate the concerns of an entire country.

Immigration

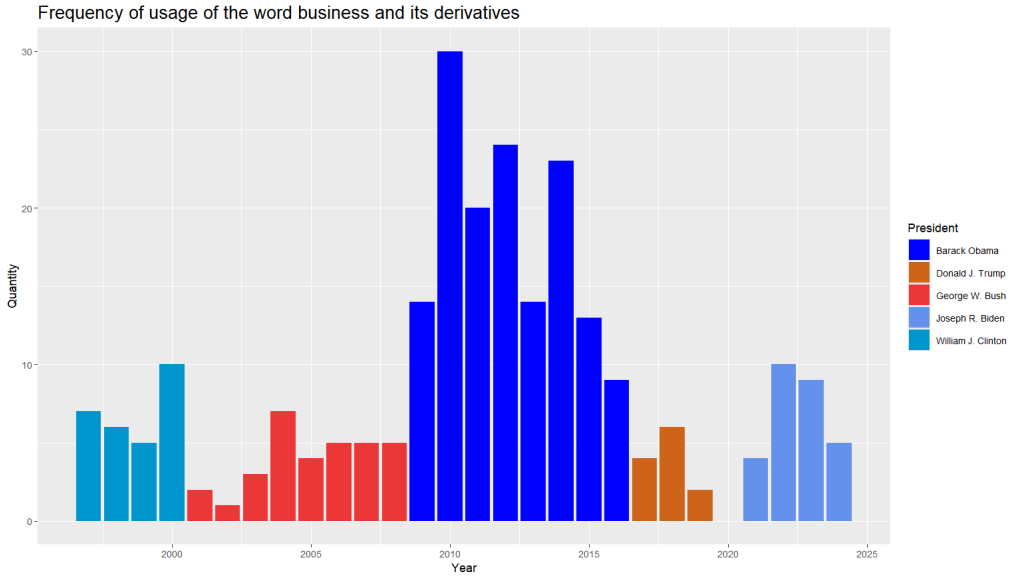
Another word that illustrates a similar issue with the data is *immigration* (and its derivatives found with regular expression "*immigra[\\w]{1,}*").

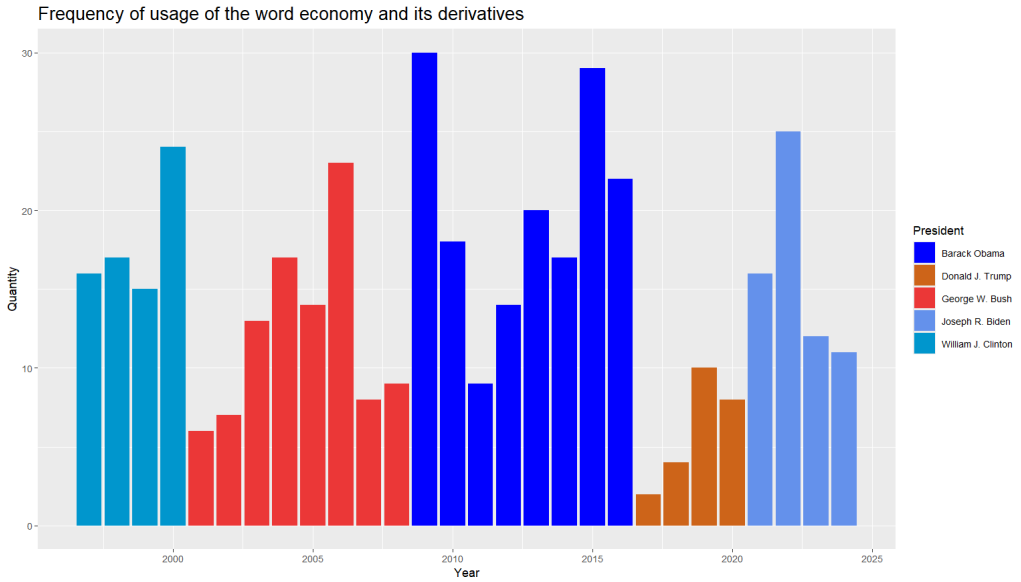


While immigration might have become a more pressing issue for Americans in 2017, a more likely explanation is that the sudden increase in popularity of the word is Donald Trump's responsibility.

Economy, business

Accepting that the data reflects the views of people delivering the speeches, I decided to consider words related to national politics - economy (regular expression: *"econom[\\w]{1,}"*) and business (regular expression: *"business[\\w]{0,}"*). I have created two plots for clarity.

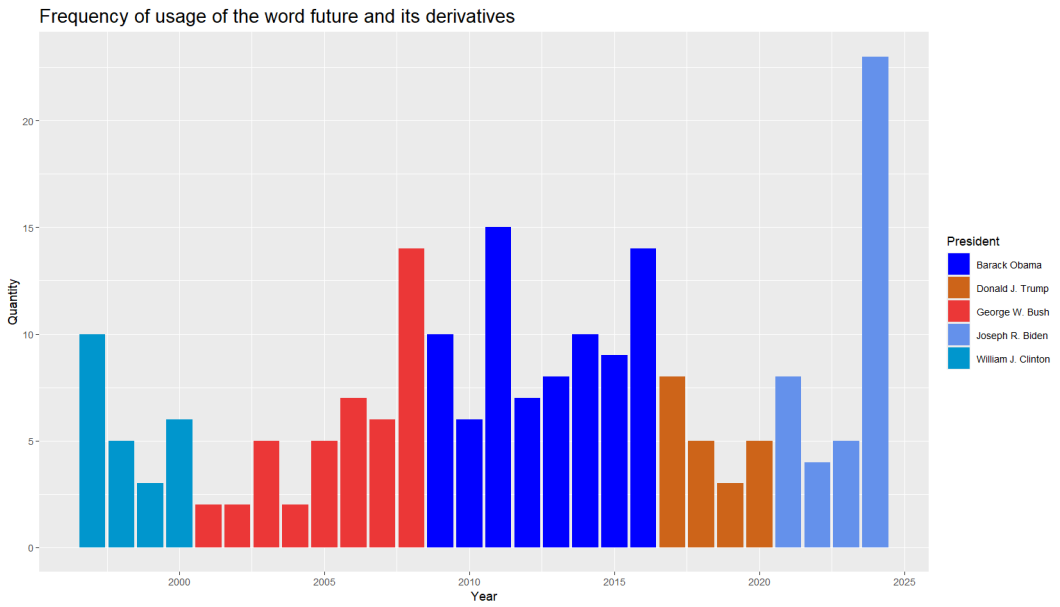




They both show Donald Trump's disinterest in the topic, while also highlighting Barack Obama's passion for talking about business.

Future

Another word that appears in every speech is *future* (regular expression: *"future[\\w]{0,}"*).



A significant increase in the popularity of the word can be noticed in 2024, possibly pointing to American's anxiety - or excitement - about the future.

Conclusions

While it is possible to make some observations about the concerns of the American people by looking at the contents of Addresses on the State of the Union, this is not the perfect data to do that. The content of the speech is heavily influenced by the political affiliation and personal preferences of the person delivering it, which makes it a less reliable data source.

Most of the observations seem quite intuitive (*terror* becoming more frequent after the 2001 attacks; *health* becoming a more popular word in the year a healthcare reform is proposed; Donald Trump focusing more on immigration-related issues, etc.), but some trends, such as the time-related anxieties (*future*; *century*) could be researched further. The American perspective on family values (*child*, *family*) and the way it has changed over time could also be an interesting research idea.